

Kalkidan Tsegaye Gebresilassie

ELECTROMECHANICAL ENGINEER | DATA ANALYST | MACHINE LEARNING ENGINEER

☎ +2519-2701-9495 ✉ kalkidantsegaye171@gmail.com 🔗 www.linkedin.com/in/kalkidan-tsegaye

📍 Addis Ababa, Ethiopia

EDUCATION & CERTIFICATIONS

Bachelor of Science in Electromechanical Engineering

Year of Graduation: June 29, 2022

Institution: Hawassa University

Final CGPA: 3.55/4.0

Completed a comprehensive engineering program integrating mechanical, electrical, and control systems. Gained strong skills in system design, automation, manufacturing processes, and applied engineering analysis. Demonstrated consistent academic excellence with a high CGPA and hands-on experience through projects, laboratory work, and practical problem-solving.

SKILLS

- **Programming & Data Technologies:** Python, C++, MATLAB / Data Analysis & Visualization (Pandas, NumPy, Matplotlib) / Machine Learning Algorithms (Regression, Classification, SVM, Model Evaluation) / SQL Basics & Data Cleaning Techniques
- **Engineering & Design Tools:** Mechanical Design (SolidWorks, CATIA, AutoCAD) / Electrical Design & Simulation (Proteus, MATLAB/Simulink)
- **Research & Development Skills:** Prototype development and performance analysis / Documentation, technical reporting, and experimental setup
- **Core Engineering Abilities:** Problem-solving & analytical thinking / Engineering calculations, FEA basics, and troubleshooting / Mechatronics system integration / Strong understanding of electromechanical components and systems
- **Soft Skills:** Research & technical documentation / Team collaboration and communication / Adaptability and continuous learning

PROFESSIONAL EXPERIENCE

Green Tech Africa

June 2023 to present

Research and Development Assistance

- **Design and Prototyping:** Collaborate with the research team to design and develop prototypes by creating models, fabricating or assembling components, and troubleshooting technical issues.
- **Data Analysis:** Assist in analyzing research data by processing experimental data, performing calculations, generating graphs or visualizations, and interpreting the results.
- **Documentation and Reporting:** Maintain accurate records of research activities, experimental procedures, and results. Contribute to the preparation of research reports, technical papers, and presentations
- **Collaboration and Communication:** Communicate research progress, challenges, and findings effectively, both verbally and in written form.

Brand engineering

Feb 2023 to May 2023

Electromechanical designer

- **System Design:** HVAC, electrical power distribution, lighting, plumbing, fire protection, and other related systems.
- **Technical Drawings and Documentation:** Prepare detailed technical drawings, schematics, and specifications using computer-aided design (CAD) software,
- **Cost Estimation:** Assist in cost estimation and budgeting for electromechanical systems.

MOHA Softdrinks Industry

June 2021 to Sep 2021

Electromechanical engineering intern

- **Responsible for controlling machines**

VOLUNTEER

Electromechanical Club – Hawassa University

Mar 2019 – Jun 2022

Coordinator

- Organized technical workshops, engineering competitions, and student innovation events
- Supported students in developing engineering projects and technical problem-solving
- Facilitated collaboration between members and instructors.

Niber Foundation

Nov 2023 – Present

Member

- Contributing to community initiatives aimed at supporting and empowering war-affected populations
- Assisting in organizing outreach programs, resource distribution, and awareness activities
- Participating in planning sessions to improve community recovery and development efforts

AWARD

Academic Excellence Recognition

Jan 2020

Hawassa University Institute of Technology

Award presented by: Dr. Nigatu Wondrade and W/ro Meskerem Estifanos

- Recognized for outstanding academic performance among female engineering students

Semester Project Completion Award

Feb 02, 2020

Hawassa University Institute of Technology

Award presented by: Mr. Demoz Lisanework

- Acknowledged for successful completion and presentation of an engineering semester project
- Designed and developed an object-tracking robot using Raspberry Pi
- Demonstrated the project during a technological exhibition showcasing innovative student research

CERTIFICATE

Machine Learning Specialization

Mar 7, 2024

DeepLearning.AI (Instructor: Andrew Ng)

Verified Certificate ID: <https://coursera.org/verify/specialization/LAYQT9ELGX96>

- Completed all 3 courses: Supervised Machine Learning: Regression & Classification / Advanced Learning Algorithms / Unsupervised Learning, Recommenders & Reinforcement Learning
- Gained practical skills in supervised and unsupervised ML, neural networks, decision trees, clustering, anomaly detection, recommender systems, and basic reinforcement learning.

Introduction to Computer Science and Programming Using Python

Mar 30, 2024

MITx – Massachusetts Institute of Technology

Verified Certificate ID: <https://courses.edx.org/certificates/7b7bcb417726426e90e951b928f3382a>

- Learned Python programming fundamentals, algorithms, computational thinking, and problem-solving.

Machine Learning Internship (1 Month)

Jan 15 – Feb 15, 2024

Prodigy InfoTech

- Completed a hands-on internship with outstanding remarks; received a Certificate of Completion and Recommendation Letter.
- Worked on model development, data preprocessing, and evaluation.

Projects:

- House Price Prediction using Linear Regression
- Customer Segmentation with K-Means Clustering
- Hand Gesture Recognition Model
- Cat vs. Dog Image Classification using SVM

REFERENCES

Mr. Beselamu Godebo

Lecturer (thesis project advisor)
Hawassa University
Gmail: Beselamu.g@gmail.com
Phone: (+251) 910834912

Mr. Daniel Dilbie Tessema

Project Manager
Green Tech Africa
Phone: (+251) 91338 9514

Mr. Demoz Lisanework

Electyromechanical engineering
department head
Hawassa University
Gmail: demoz0504@gmail.com
Phone: (+251) 915975247

Mr. Anteneh Taye Bekele

Operation Manager
Green Tech Africa
Phone: (+251) 91 162 3999